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- Spatial distribution of mirror and AIC waves in the magnetosheath
- Dependence of magnetosheath occurrence mirror and AIC waves on shock parameters
- Competition of mirror and ion cyclotron instabilities in the magnetosheath

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Magnetosheath plasma stability and ULF wave occurrence as a function of location in the magnetosheath and upstream bow shock parameters

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Abstract We present the results of a statistical study of the distribution of mirror and Alfvén-ion cyclotron (AIC) waves in the magnetosheath together with plasma parameters important for the stability of ULF waves, specifically ion temperature anisotropy and ion beta. Magnetosheath crossings registered by Cluster spacecraft over the course of 2 years served as a basis for the statistics. For each observation we used bow shock, magnetopause, and magnetosheath flow models to identify the relative position of the spacecraft with respect to magnetosheath boundaries and local properties of the upstream shock crossing. A strong dependence of both plasma parameters and mirror/AIC wave occurrence on upstream Θ_{Bn} and M_A is identified. We analyzed a joint dependence of the same parameters on Θ_{Bn} and fractional distance between shock and magnetopause, zenith angle, and length of the flow line. Finally, the occurrence of mirror and AIC modes was compared against the respective instability thresholds. We noted that AIC waves occurred nearly exclusively under mirror stable conditions. This is interpreted in terms of different characters of nonlinear saturation of the two modes.

1. Introduction

In this article we address the problem of spatial distribution of ultralow frequency (ULF) waves and plasma parameters determining the stability of plasma in the terrestrial magnetosheath. Based on a large statistical data set, we studied the relative occurrence of the waves in relation to the parameters of the upstream shock crossing and the location of the observation within the magnetosheath.

Already, early satellite observations have shown that the magnetosheath is populated by intense magnetic field fluctuations [*Fairfield and Ness*, 1970] at time scales from 1 s to tens of seconds and explanations of their origin in terms of local plasma instabilities were proposed already in that era [*Kaufmann et al.*, 1970; *Fairfield*, 1976]. The ion instabilities responsible for these oscillations are driven by ion temperature anisotropy $A = T_\perp/T_\parallel > 1$ in high ion beta plasma $\beta_\parallel = 2\mu_0 nkT_\parallel/|\mathbf{B}|^2 > 1$, conditions typical for the magnetosheath [*Sckopke et al.*, 1990; *Thomsen et al.*, 1985]. Here n denotes plasma ion density, $\mathbf{B}$ the magnetic field vector, and subscripts $\parallel$ and $\perp$ correspond to components parallel and perpendicular to local $\mathbf{B}$.

Later research based primarily on data from ISEE and AMPTE satellites has shown that two wave modes constitute a large majority of magnetosheath ULF waves: the mirror mode waves and kinetic Alfvén-ion cyclotron (AIC) waves, also named electromagnetic ion cyclotron waves in the literature [*Anderson and Fuselier*, 1993; *Anderson et al.*, 1994; *Denton et al.*, 1994; *Song et al.*, 1994; *Hubert et al.*, 1989, 1998; *Lacombe et al.*, 1992]. For a more recent review of magnetosheath wave modes, see *Schwartz et al.* [1996].

The mirror modes are generated locally by the mirror instability [*Hasegawa*, 1969; *Hellinger*, 2007] when its threshold condition is satisfied. In the simplest approximation of proton-electron plasma with cold electrons the instability condition can be expressed as

$$R = \beta_\perp (A - 1) > 1. \tag{1}$$

The parameter R defined here is the mirror instability parameter used throughout this study to evaluate stability of plasma with respect to mirror instability.

The ion cyclotron instability [*Davidson and Ogden*, 1975] responsible for the generation of AIC waves often grows under the same conditions as the mirror instability and in the linear approximation should dominate in

lower beta plasmas [*Gary*, 1992; *Gary et al.*, 1993]. The competition between the two instabilities is, however, a more complex problem [*Shoji et al.*, 2009] and will be discussed later in this article.

It was demonstrated that the above instabilities are important for the global dynamics of magnetosheath plasmas: they dissipate the excess temperature anisotropy created by shock heating and keep magnetosheath plasmas at the state of marginal stability [*Fuselier et al.*, 1994; *Hellinger et al.*, 2003].

The spatial distribution of the ULF waves in the magnetosheath has also been studied previously. It was shown that the AIC waves are found predominantly near the bow shock [*Song et al.*, 1994; *Hubert et al.*, 1998] and close to the magnetopause and, in particular, in the plasma depletion layer [*Anderson and Fuselier*, 1993], while mirror structures dominate in the central and downstream magnetosheath [*Génot et al.*, 2009a; *Verigin et al.*, 2006] but can occur immediately downstream from the shock too [*Hubert et al.*, 1989]. The above studies also showed that the ULF fluctuations are generally stronger in the dayside magnetosheath. *Anderson and Fuselier* [1993] performed a small-scale statistical study of the dependence of the observed ULF waves on the upstream shock type. They distinguished between quasi-perpendicular (Q⊥) and quasi-parallel (Q||) magnetosheath based on a flux of locally observed energetic He^{++} ions which is known to be stronger behind Q|| shock. This coarse technique allowed to establish that mostly AIC waves were observed in Q|| magnetosheath, while in the Q⊥ magnetosheath the ratio of mirror and AIC observations was close to 1. In their data set, the occurrence rate of either type of wave was lower in Q|| magnetosheath.

The mirror structures have received considerable attention recently, when their nonlinear behavior and shape have been addressed. Mirror structures are known to appear in several forms: mirror peaks, dips, and sometimes more irregular or quasi-sinusoidal structures. Several statistical and numerical studies described the relative occurrence of peaks and dips as a function of local plasma parameters and location within magnetosheath [*Soucek et al.*, 2008; *Califano et al.*, 2008; *Génot et al.*, 2009b, 2011]. It was shown that mirror peaks only appear in mirror unstable plasma ($R > 1$), most often in central dayside magnetosheath, while dips can exist for even lower values of R and dominate in low-β plasma, such as in the magnetosheath flanks and close to the magnetopause.

The character of locally generated ULF waves is determined by local plasma properties which are largely affected by the process of heating and compression of solar wind plasma taking place at the bow shock. From Rankine-Hugoniot relations [*Chao et al.*, 1995; *Génot*, 2009] it can be shown that downstream temperature anisotropy and β are determined by solar wind parameters and the local properties of the shock: the angle Θ_{Bn} between the shock normal and upstream interplanetary magnetic field (IMF) and local shock Mach number (in this work, we used Alfvénic Mach number M_A). This is true for magnetosheath plasma near the bow shock; further downstream in the magnetosheath, other effects such as expansions or instabilities affect the plasma conditions.

This dependence naturally results in an asymmetric distribution of plasma parameters and waves in the magnetosheath driven by the IMF direction and can be identified in statistical studies as dawn-dusk asymmetry due to the typical Parker spiral orientation of IMF. A number of studies summarized, for example, in a recent review article by *Walsh et al.* [2014] have shown dawn-dusk asymmetries in plasma density, velocity, magnetic field magnitude, and in the nature of magnetic field fluctuations. The dawn-dusk asymmetry dependent on the IMF orientation is conveniently characterized in the magnetosheath interplanetary medium (MIPM) reference frame [*Verigin et al.*, 2006], where the dawnside is always connected to a *Q*|| shock and duskside to Q⊥.

In the MIPM frame *Dimmock and Nykyri* [2013] showed that the dawn magnetosheath contains stronger magnetic field and faster plasma flow, but no significant asymmetry in density was found in contrast to other studies such as *Paularena et al.* [2001] and *Walsh et al.* [2012]. Statistical studies have shown that the overall level of magnetic field oscillations is larger on the Q|| sides of the magnetosheath [*Luhmann et al.*, 1986; *Dimmock et al.*, 2014] and that there is a qualitative difference in the nature of oscillations [*Shevyrev et al.*, 2007; *Petrinec*, 2013]. On the Q|| side, the field direction fluctuates on the time scale of minutes, corresponding to turbulence originating from the Q|| shock and possibly foreshock. The Q⊥ side is populated to a larger extent by smaller-scale oscillation of compressional nature, corresponding to locally generated waves.

In the present article we used a data set of 2 years of Cluster magnetosheath observations to investigate in detail the dependence of the plasma parameters relevant to the mirror and ion cyclotron instabilities

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(anisotropy A, $\beta_{\parallel}$, and mirror instability parameter R) as well as the occurrence of the ULF waves as a function of location in the magnetosheath and local parameters of the bow shock.

2. Data Set and Data Processing

The present work is based on a statistical data set of in situ plasma and field measurements performed in the magnetosheath by Cluster spacecraft [*Escoubet et al.*, 2001]. The data set used in this study covers two years of magnetosheath observations. All intervals when Cluster 1 spacecraft was located in the magnetosheath between dates 4 November 2006 and 10 June 2008 are included. These two years were chosen because of a favorable Cluster orbit: during this interval, the apogee was drifting south of the ecliptic plane and the orbital plane was rotating toward equatorial orientation resulting in a good coverage of the subsolar magnetosheath. Magnetosheath interval selection was performed by interactive visual identification of bow shock and magnetosheath crossings based on magnetic field magnitude and ion energy spectra. Two Cluster data sets were used in this study—magnetic field intensity sampled at five vectors per second from the FGM magnetometer [*Balogh et al.*, 2001] and onboard calculated ion moments (density, flow velocity, and perpendicular and parallel temperatures) from the Cluster Ion Spectrometry-Hot Ion Analyzer ion spectrometer [*Rème et al.*, 1997]. This instrument does not separate individual ion species; therefore, the obtained ion moments are an approximation of H^+ contaminated by the presence of He^{++}. In a majority of cases, the He^{++} abundance is low and the moments can be considered a good approximation of H^+ moments. A study dedicated to the He^{++} population is beyond the scope of this paper, but it would represent a valuable complement to our results. All Cluster data were obtained from the public Cluster Science Archive.

The magnetosheath data were split in 5 min long intervals, and a set of field and plasma parameters was calculated on each 5 min interval. This length of interval was chosen with regard to the characteristic scale of mirror and ion cyclotron waves whose frequencies are typically between 0.05 Hz and 1 Hz. For each interval, we calculated average values of ion moments, field magnitude, and derived quantities such as ion beta parameter and temperature anisotropy. Intervals with missing data, low-quality data, and intervals where the CIS instrument was not in a mode appropriate for magnetosheath plasma measurements were removed. Unless specified otherwise, all data presented in this paper are calculated as averages over 5 min intervals and one 5 min interval constitutes one data point in the statistics. This way we obtained a set of 26,948 data points. The distribution of the data points in GSE coordinates with respect to average magnetosheath boundaries are shown in Figure 1. The plot demonstrates a good coverage of dayside magnetosheath near the equatorial plane and in the Southern Hemisphere.

Each interval was scanned for the presence of ULF waves using an automated technique based on variance analysis of magnetic field. Previous research discussed in section 1 has shown that two wave modes are responsible for a large majority of magnetosheath ULF waves: the mirror mode waves and Alfvén-ion cyclotron waves. In observed magnetic field data, one can distinguish between these two modes relatively easily based on their different polarization [*Krauss-Varban et al.*, 1994; *Song et al.*, 1994]. Magnetic field variations corresponding to mirror waves are typically linearly polarized with the variation δB nearly parallel to background field $\mathbf{B}$. Ion cyclotron waves, on the other hand, are known to be transversely left-hand polarized with the $\mathbf{k}$ vector directed obliquely or nearly parallel to the background field [*Remya et al.*, 2014].

Each interval of data is analyzed using standard variance analysis [*Sonnerup and Sheible*, 1998] to obtain the minimum and maximum variance direction vectors. We also transform the magnetic field waveform to field-aligned coordinates (using mean field vector $\bar{\mathbf{B}}$ to obtain RMS values of fluctuations in $\bar{B}$ parallel and perpendicular direction: $\delta B_{\parallel}^2$ and $\delta B_{\perp}^2$. We now classify as mirror events those intervals, where $\delta B_{\parallel}^2/\bar{B}^2 > 0.1$, $\delta B_{\parallel}^2/\delta B_{\perp}^2 > 2$, and the angle between $\bar{\mathbf{B}}$ and the maximum variance direction is less than 25°. This criterion is similar to that used, for example, by *Génot et al.* [2009b]. Analogously, intervals are identified as AIC events when $\delta B_{\perp}^2/\bar{B}^2 > 0.1$, $\delta B_{\parallel}^2/\delta B_{\perp}^2 < 0.5$, and the angle between $\bar{\mathbf{B}}$ and the minimum variance direction is less than 45°. The mirror intervals are further classified as mirror peaks or mirror dips based on the value of normalized skewness S of the magnetic field magnitude $|\mathbf{B}|$ [*Génot et al.*, 2009b], providing a simple and easily reproducible criterion. Mirror intervals where $S > 0.1$ are classified as peaks and intervals where $S < -0.1$ as dips.

It should be noted that this simple method suitable for large-scale studies does not identify other transverse wave modes that might be present (such as the fast MHD mode), and these are classified as AIC waves. Mixed events containing a signature of both mirror and AIC waves are not resolved either.

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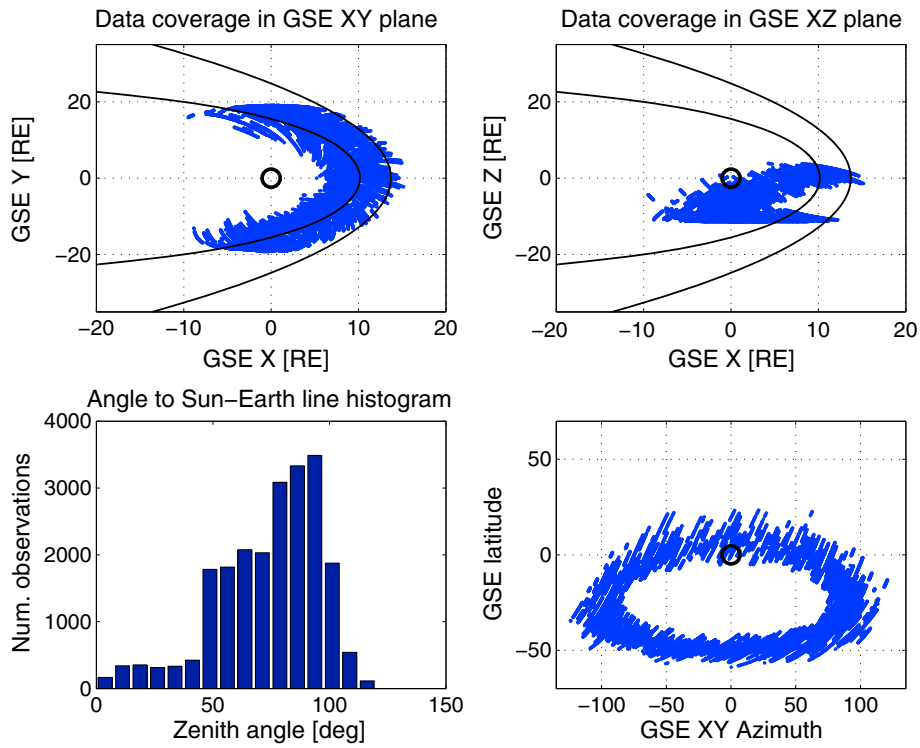

Figure 1. Coverage of the dayside magnetosheath by the data set used in this study. Each dot represents one 5 min interval of magnetosheath data. (bottom left) The zenith angle which is the angle between GSE position vector $\vec{r}$ and GSE X axis.

Similar techniques based on magnetic field were previously used to identify magnetosheath mirror waves [*Génot et al.*, 2009b] or both mirror and ion cyclotron waves but in spectral domain [*Anderson et al.*, 1994; *Denton et al.*, 1994; *Song et al.*, 1994].

2.1. Estimating Position in the Magnetosheath and Corresponding Bow Shock Parameters

One of the objectives of our study is to relate the local properties of waves and plasmas to the parameters of the bow shock. Previous studies discussed above in section 1 have shown that a significant dawn-dusk asymmetry exists in magnetosheath parameters and waves and that it is most likely due to the asymmetry introduced by IMF orientation and the location of Q⊥ and Q|| shocks.

Clearly, the local parameters of the shock, in particular the angle Θ_{Bn}, vary very strongly with the location on the bow shock surface. In this paper we take a direct approach to the problem and attempt to derive for each magnetosheath observation the parameters of the bow shock at the location and time where the observed plasma element entered the magnetosheath. Our method employs backtracing the magnetosheath plasma flow line from the observation point to the shock to obtain the probable entry point of the observed plasma into the magnetosheath. A similar approach was employed previously for tracing the evolution of mirror structures [*Tátrallyay et al.*, 2008; *Génot et al.*, 2011], for magnetosheath ion fluxes [*Hayosh et al.*, 2005] and magnetic field oscillations [*Luhmann et al.*, 1986; *Shevyrev and Zastenker*, 2005].

To perform this tracing procedure, we used a magnetosheath flow model described and validated in *Soucek and Escoubet* [2012]. In the present study we also used the same dynamic bow shock and magnetopause models with OMNI2 data set providing the upstream solar wind parameters: the *Farris et al.* [1991] model for the bow shock and *Shue et al.* [1997] for the magnetopause. Correction for aberration of solar wind by the orbital motion of the Earth was consistently applied.

This flow line tracing was performed for every point in the data set, and for each point we obtained the corresponding point of origin on the model bow shock and the time required for the observed plasma element to be convected from the shock to the spacecraft location. Using those two parameters and the same *Farris et al.* [1991] bow shock model, we then calculated the local parameters of the source shock: the Alfvén Mach number M_A and the angle Θ_{Bn} between model shock normal and IMF. This method assumes that the position of

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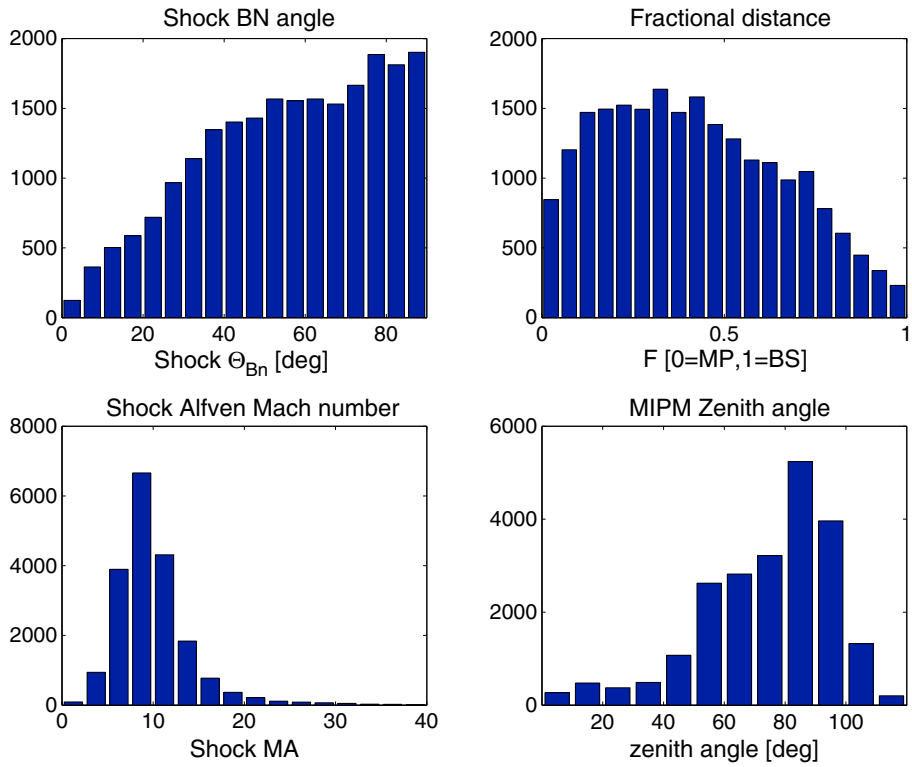

Figure 2. A distribution of traced shock parameters, Θ_{Bn} and M_A, and parameters describing the location of the spacecraft within the magnetosheath F and ϑ.

the bow shock did not change very much during the time needed for the plasma to propagate from the shock to spacecraft location (median propagation time in our data set is 180 s). The method is less sensitive to rapid IMF changes as the OMNI2 magnetic field used in M_A and Θ_{Bn} calculation was adjusted for the propagation time and the position of the dayside bow shock is only weakly dependent on the IMF orientation.

For each interval we also calculated the relative position of the observation point within model magnetosheath boundaries expressed by fractional distance F and zenith angle ϑ defined as in *Verigin et al.* [2006]:

$$F(\mathbf{r}) = \left(|\mathbf{r}| - r_{mp}\right) / \left(r_{bs} - r_{mp}\right) \tag{2}$$

$$\vartheta(\mathbf{r}) = \arccos\left(\left(\mathbf{r}, \mathbf{e}_x\right) / |\mathbf{r}|\right). \tag{3}$$

Here $\mathbf{r}$ is the spacecraft position vector in GSE coordinates, $\mathbf{e}_x$ the unit vector in the aberrated GSE X direction, and r_{bs} and r_{mp} are the geocentric distances of the model bow shock and magnetopause in the direction given by zenith angle ϑ (as the bow shock magnetopause models are cylindrically symmetric, these distances are fully determined by ϑ). For more details refer to *Soucek and Escoubet* [2012].

Histograms of the estimated parameters describing magnetosheath position and traced bow shock crossing for the entire data set are plotted in Figure 2. Apparently, the coverage of the parameter space is good. The subsolar magnetosheath ($\vartheta < 40°$) is less represented due to its lower volume and the quasi-parallel shock with low $\Theta_{Bn} < 30°$ is encountered less frequently in accordance with the lower probability of encountering such a shock in nature.

3. Statistical Properties of Plasma Parameters and ULF Wave Occurrence

The most important parameters controlling the stability of magnetosheath ULF waves in the magnetosheath are the ion temperature anisotropy A, ion $\beta_{||}$, and mirror instability parameter R. In this section we will use the

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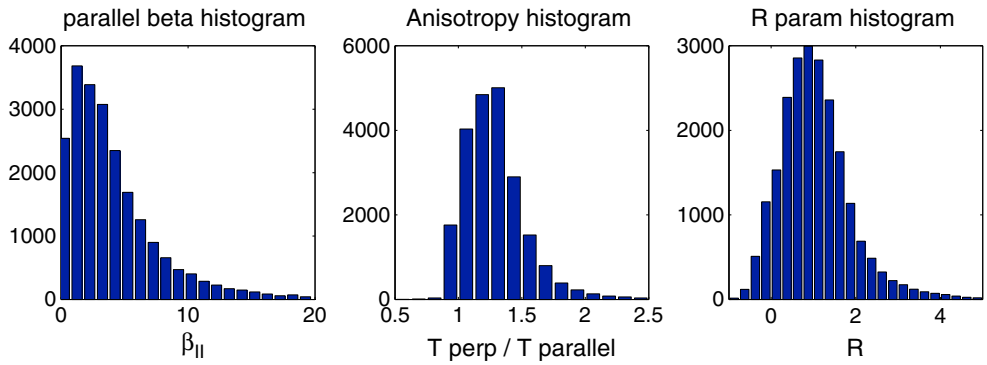

Figure 3. Histograms of ion temperature anisotropy, parallel ion beta, and the mirror instability parameter *R* in the analyzed data set.

statistical data set introduced in the previous section to characterize the distribution of these parameters in the magnetosheath together with the relative occurrence of mirror and AIC waves.

While many authors have discussed these parameters, a systematic statistical characterization has not been performed to date and is required for interpretation of the ULF wave observations. The overall histograms of the three parameters are given in Figure 3, with median $A = 1.26$, median $\beta_{||} = 3.43$, and median $R = 0.96$. This figure is included to provide a context to the more detailed analysis, and the values are consistent with the typical values reported by previous works [*Phan et al.*, 1994; *Hill et al.*, 1995].

3.1. Dependence of Plasma Stability and Wave Occurrence on Bow Shock Parameters

The dependence of the investigated plasma parameters on the properties of the upstream shock crossing is shown in Figure 4. Color coded are median values of *A*, ion $\beta_{||}$, and *R* calculated from data points binned by Θ_{Bn} and M_A. The strong dependence of all three plasma parameters on the shock properties is evident: the strongest anisotropy is observed at quasi-perpendicular shocks with low Mach number, while quasi-parallel shocks with large Mach number are correlated with the highest values of $\beta_{||}$. The observed anisotropy-beta anticorrelation can be attributed to the double adiabatic fluid theory Chew-Goldberger-Low (CGL) law which predicts an inverse dependence between *A* and $\beta_{||}$ during adiabatic plasma compression or expansion. CGL law can be applied to magnetosheath plasma expansion on the global scale, but both mirror and ion cyclotron instabilities are known to constrain this relationship by dissipating excess anisotropy [*Anderson and Fuselier*, 1993; *Fuselier et al.*, 1994].

Figure 4d shows the number of values in each bin used to calculate the medians and demonstrates a good statistical coverage with the exception of rare nearly parallel high Mach number shocks. The mirror instability parameter *R* indicates that the most unstable conditions occur for quasi-perpendicular high Mach number shocks. These dependencies are qualitatively consistent with the trends calculated from anisotropic Rankine-Hugoniot conditions [*Génot*, 2009; *Chao et al.*, 1995], but a direct comparison to the theory is impossible without additional information on the shock properties not available in our model. Later in this article we show that other factors contribute to this statistics, but the dominant trends are those presented here.

In Figure 5 is presented a statistics of ULF wave occurrence as a function of the same shock parameters as in Figure 4. Here the color represents the relative occurrence of each type of ULF wave: the ratio of the number of 5 min intervals with identified waves of a specific type to the total number of intervals in the data set falling into this bin. This figure needs to be interpreted together with Figure 4 as the wave occurrence depends strongly on the average plasma parameters. The occurrence of identified mirror structures is plotted for all identified mirror events (Figure 5c), for mirror peaks defined by positive skewness $S > 0.1$ (Figure 5a), and mirror dips with negative skewness $S < -0.1$ (Figure 5b). Mirror peaks are most frequently observed for high Mach number Q⊥ shock, in the same region where the *R* parameter (Figure 4b) is the largest. Mirror dips, on the other hand, are distributed more evenly in the parameter space but favor quasi-perpendicular conditions and appear within the region of lower $\beta_{||}$ and moderate Mach numbers as seen in Figure 4c. The above dependence correlations between *R*, $\beta_{||}$, and mirror wave occurrence agrees well with previous results [*Soucek et al.*, 2008], the dependence on M_A with results of *Génot et al.* [2009a], and Θ_{Bn} coarsely agrees with the MIPM frame result of *Verigin et al.* [2006]. Figure 4d shows the distribution of identified AIC waves.

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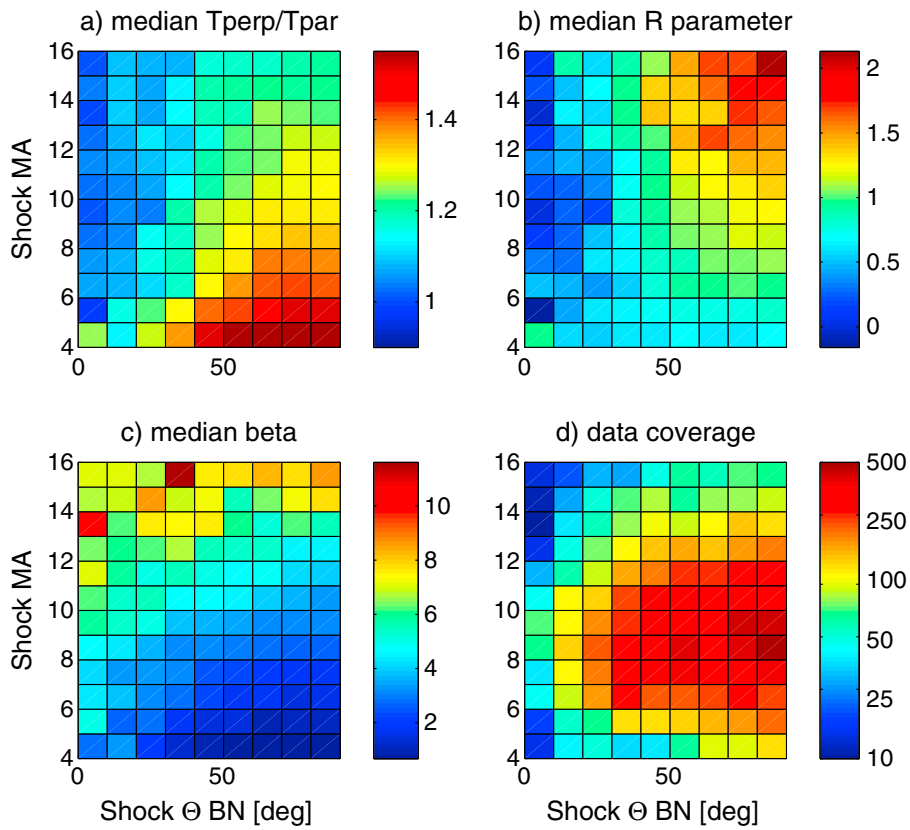

Figure 4. (a) Median values of ion temperature anisotropy A, (c) parallel ion beta and (b) the mirror instability parameter R calculated in two-dimensional bins of corresponding shock parameters Θ_{Bn} and M_A. (d) The number of data points in each bin.

Interestingly, these waves are clustered in the low-M_A part of the histogram for moderate Q⊥ conditions with most values between $50° < \Theta_{Bn} < 80°$. The observation that significantly more mirror waves are observed in Q⊥ magnetosheath corresponds well with *Anderson and Fuselier* [1993], but contrary to their statistics, our results show a lack of AIC observation in the Q|| part. It can be seen that the region of parameter space occupied by AIC waves corresponds to low-$\beta_{||}$ plasmas and that AIC waves were found exactly for those shock conditions not favorable for mirror modes. The competition between mirror and ion cyclotron instabilities [*Gary et al.*, 1993] is most likely responsible for this effect. As seen in Figure 4, low Mach number shocks result in highly anisotropic, low-β plasmas. Under these conditions the mirror instability is usually inhibited ($R < 1$) and the AIC waves are more likely to be observed (cf. also section 3.3).

3.2. Dependence on Θ_{Bn} and Location in the Magnetosheath

In Figures 6 and 7 is presented a joint statistical dependence of the studied plasma parameters and wave occurrence rates on the angle Θ_{Bn} and on coordinates, characterizing the location of the observation with respect to magnetosheath boundaries, specifically on magnetosheath fractional distance F, zenith angle ϑ, and on the length of the traced plasma flow line l_F from the observation point to the model bow shock. The value of l_F is calculated from the magnetosheath flow model and expressed in units of Earth radius R_E.

The color-coded median values of A, $\beta_{||}$, and R and the relative occurrences of individual ULF wave types are defined analogously to the previous section. Each panel in Figure 6 shows a median calculated in two-dimensional bins of Θ_{Bn} and one of the parameters F, ϑ, and l_F.

The temperature anisotropy (Figures 6a–6c) is clearly higher in the quasi-perpendicular magnetosheath as observed already in the previous section but is also stronger in the dayside magnetosheath ($\vartheta < 50°$) and in the inner half of the magnetosheath ($F < 0.5$). Close to the magnetopause, the dependence of median anisotropy on Θ_{Bn} is weaker than close to the bow shock and higher values are observed for $\Theta_{Bn} \geq 40°$. This suggests that temperature anisotropy is driven by a combination of two effects: the shock heating and expansion near

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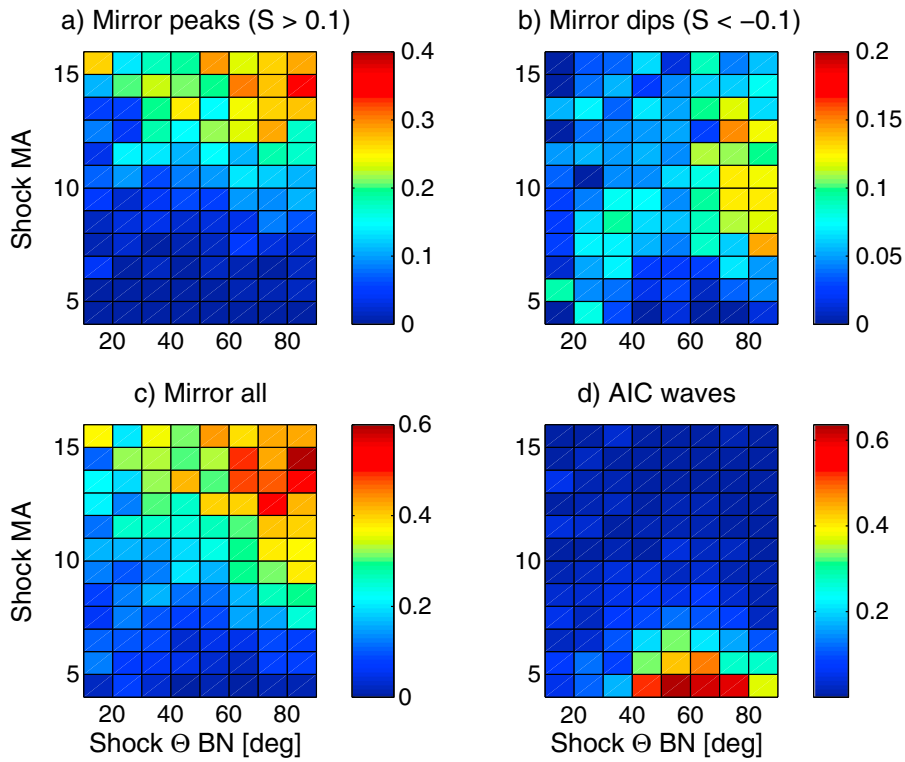

Figure 5. A relative occurrence of (a) mirror peaks, (b) mirror dips, (c) all mirror waves, and (d) AIC waves in two-dimensional bins of corresponding shock parameters Θ_{Bn} and M_A. Relative occurrence is defined as the number of intervals containing the corresponding wave type to the total number of intervals within each bin.

the magnetopause and specifically in the plasma depletion layer [*Anderson and Fuselier*, 1993], resulting in larger values of anisotropy according to double adiabatic theory. Both of these effects are weaker in the flank magnetosheath which explains the dependence on ϑ observed in Figure 6b. The dependence of anisotropy on l_F is not very strong with a small but observable decrease for large values of l_F corresponding to flank magnetosheath.

The median values of $\beta_{\|}$ (Figures 6d–6f) follow an approximately inverse pattern due to the anticorrelation between the two parameters discussed above, but some deviations from this general rule are observed. Ion beta appears to exhibit a strong dependence on Θ_{Bn} with the largest values for Q|| shocks. This dependence is very profound closer in the inner magnetosheath ($F < 0.5$) and at the flanks (Figure 6e) and for large f_L (Figure 6f), but almost nonexistent near the bow shock (Figures 6d and 6f). Figure 6e shows that low-beta plasmas are found both in the Q⊥ flank magnetosheath and in Q⊥ subsolar magnetosheath. In the subsolar magnetosheath, low $\beta_{\|}$ is associated with very high anisotropy, while in the flank magnetosheath this anti-correlation is not evident. The distribution of R clearly shows that most mirror unstable plasma is found in the subsolar quasi-perpendicular magnetosheath and that the R parameter decreases with the flow line length l_F, consistently with *Génot et al.* [2009a, 2011].

This dependence agrees with the mechanism proposed in *Soucek et al.* [2008]: in subsolar magnetosheath plasma depletion layer (PDL) is formed and the adiabatic expansion results in decreased $\beta_{\|}$ and increased anisotropy. Plasma in the flank magnetosheath generally corresponds to larger values of l_F, also associated with low beta and moderate anisotropy. The expansion was slower and the plasma instabilities had a sufficient time to decrease the anisotropy; the flank plasma is thus generally filled with low-β low-anisotropy mirror stable plasmas.

In Figure 7 we show the relative occurrence of different types of ULF waves defined and visualized analogously to section 3.1. In accordance with previous results [*Soucek et al.*, 2008; *Génot et al.*, 2009b; *Califano et al.*, 2008] mirror peaks (Figures 7a–7c) appear in a strict correlation with high values of R and predominantly in dayside, middle magnetosheath behind Q⊥ shock. Mirror dips, on the other hand, appear in both subsolar Q⊥

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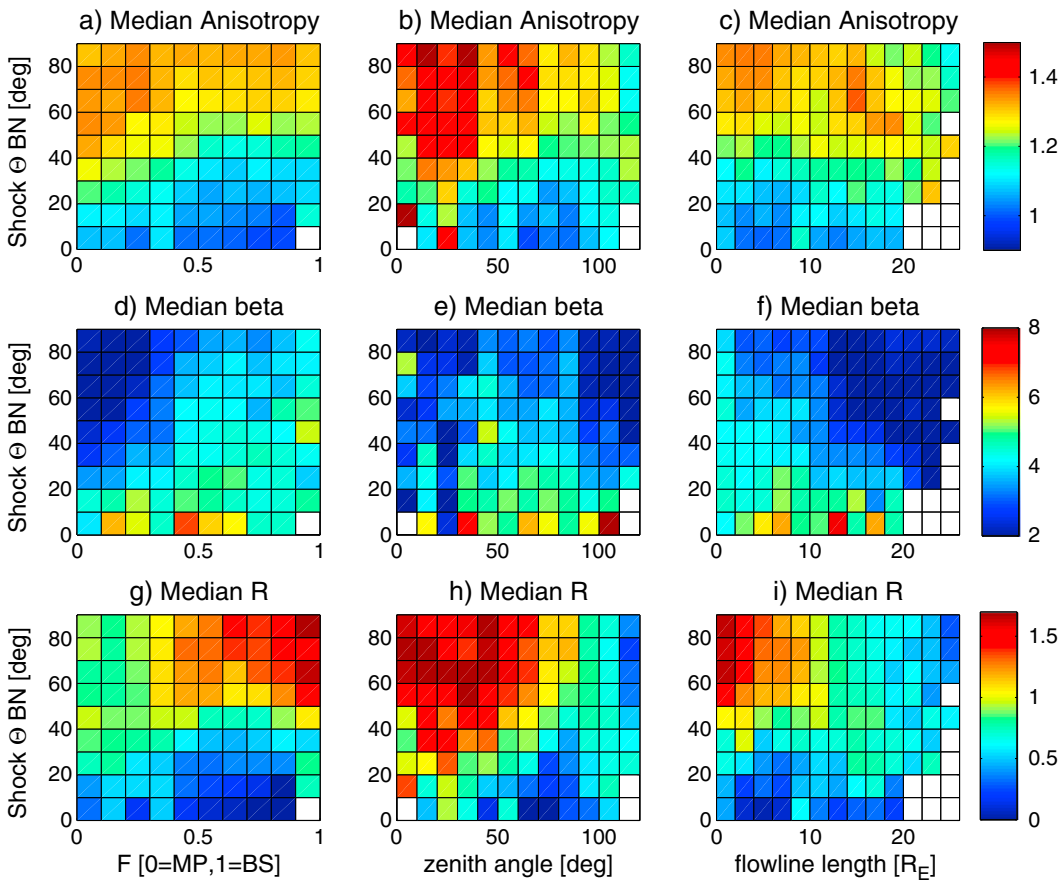

Figure 6. (a–c) Median values of ion temperature anisotropy, (d–f) parallel ion beta, and (g–i) the mirror instability parameter R calculated in two-dimensional bins of Θ_{Bn} and fractional distance F (Figures 6a, 6d, and 6g), zenith angle ϑ (Figures 6b, 6e, and 6h), and flow line length l_F (Figures 6c, 6f, and 6i).

magnetosheath close to the magnetopause (in correlation with low $\beta_{||}$) and toward the flanks for $\vartheta > 90°$. Mirror dips also show a very strong dependence on l_F for all angles of Θ_{Bn}, consistent with *Génot et al.* [2011]. The distribution of AIC waves shows a more interesting pattern. In all three bottom panels (Figures 7g–7i) we observe that AIC waves generally appear for moderate Q⊥ conditions ($45° < \Theta_{Bn} < 80°$) as already seen in Figure 5. There are two distinct clusters of AIC wave activity apparent in Figures 7g and 7h: a strong occurrence near the bow shock ($F > 0.6$) at medium zenith angles $50° < \vartheta < 90°$ and a weaker patch of AIC waves near the magnetopause, probably corresponding to PDL waves. As seen in Figure 7i, AIC waves are observed for all values of l_F with local maxima just behind the shock and for values between 12 and 20 Earth radii. The occurrence of AIC waves behind the shock was noted by *Song et al.* [1994] and *Hubert et al.* [1998], and the occurrence in PDL is well documented in *Anderson and Fuselier* [1993] and *Anderson et al.* [1994]. Similar to *Fuselier et al.* [1994], our statistics shows very few AIC waves for Q|| conditions. The occurrence rate of AIC waves in the PDL identified in our study is relatively low, possibly due to the fact that some of the waves may appear at frequencies above 2 Hz which are not resolved properly in our data.

3.3. Distribution of ULF Waves in Comparison With Instability Thresholds

In the literature, the distribution of magnetosheath waves is often plotted in anisotropy-beta space to compare the occurrence of the waves with local plasma parameters and instability thresholds [*Soucek et al.*, 2008; *Hellinger et al.*, 2003; *Anderson et al.*, 1994]. Such a plot for our data set is shown in Figure 8. In the figure, red and green dots represent the individual mirror and AIC events, respectively. Overplotted is the mirror instability threshold $R = 1$ and a fitted proton cyclotron instability threshold condition from *Hellinger et al.* [2006]: $A = 1 + 0.43/(\beta_{||} + 0.0004)^{0.42}$. In this figure, the mirror mode distribution is identical to that shown in *Soucek et al.* [2008], showing that mirror structures are distributed along the threshold curve. The AIC waves appear for lower $\beta_{||}$ as expected from the competition of the two instabilities [*Gary et al.*, 1993], but the distribution

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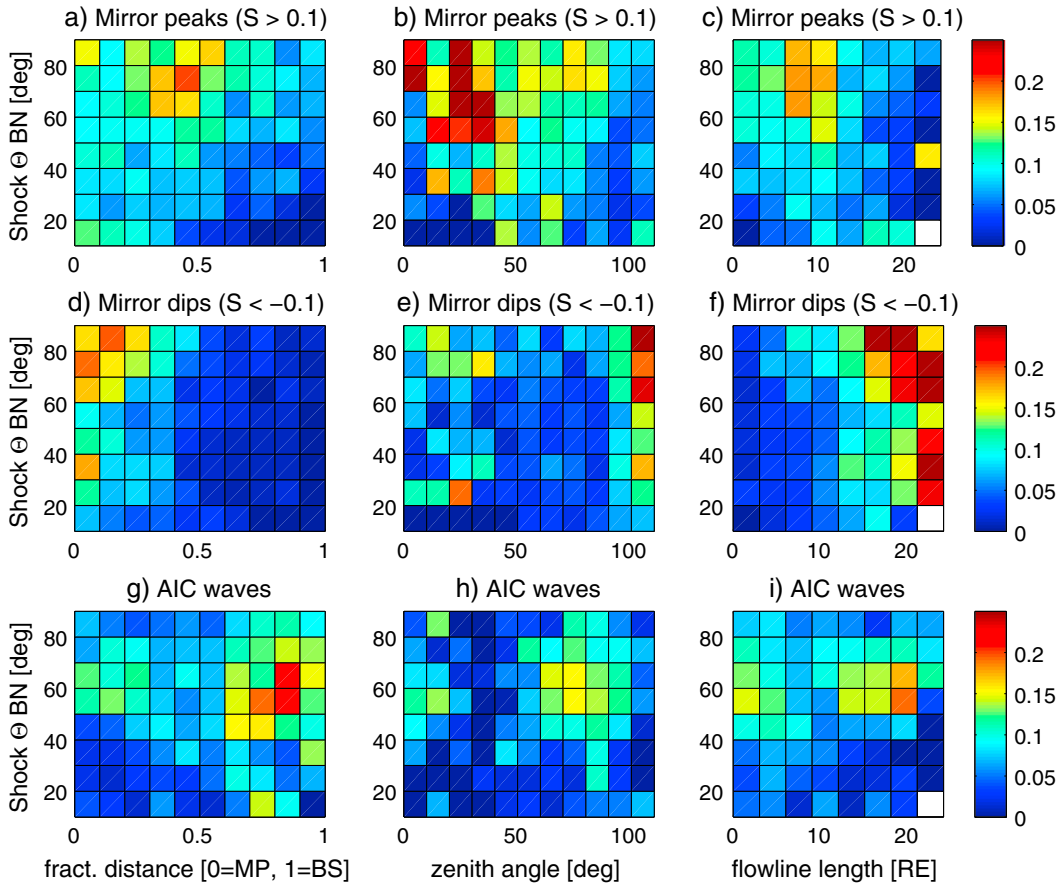

Figure 7. A relative occurrence (defined as in Figure 5) of (a–c) mirror peaks, (d–f) mirror dips, and (g–i) AIC waves in two-dimensional bins of Θ_{Bn} and fractional distance F (Figures 7a, 7d, and 7g), zenith angle ϑ (Figures 7b, 7e, and 7h) and flow line length l_F (Figures 7c, 7f, and 7i).

is relatively broad with values $\beta_{\|} \leq 5$ well represented. Similar to the case of mirror waves, the AIC waves follow the IC threshold curve but appear both above and below the threshold with a considerable scatter. The results differ in some aspect from a similar plot (with only a few data points) published by *Anderson and Fuselier* [1993] which has shown a strict division based on the value of $\beta_{\|}$.

A remarkable feature stands out in this figure: the AIC waves are strictly limited to the area below the mirror threshold curve. This can be interpreted in terms of competition between mirror and IC instabilities. AIC waves only appear when mirror instability is inhibited. This can suggest that when linear theory predicts a positive growth rate for both instabilities, the mirror instability dominates despite the fact that the growth rate of ion cyclotron waves should be larger. In the case when mirror instability threshold is not reached, the AIC waves can grow uninhibited.

A convenient explanation is provided by the different saturation properties of the instabilities as demonstrated in three-dimensional hybrid simulations by *Shoji et al.* [2009]. In their simulation, the authors follow the competition of the two instabilities from initial linear stage to nonlinear saturation. It is shown that while initially, the ion cyclotron stability grows faster, the growth is quickly saturated and the AIC waves are damped after a short time. The mirror waves, on the other hand, while initially growing slower, reach a stable saturated state in which large-amplitude $|B|$ oscillations persist for a long time. This mechanism explains this feature and also the large dominance of mirror modes over AIC waves in magnetosheath observations.

A similar effect has been analyzed by *Russell et al.* [1999] in their study of Galileo observations of ion cyclotron and mirror waves in the wake of Io. In their data set, the ion cyclotron waves at SO_2^+ gyrofrequency were observed during the Io approach, but two narrow bands of mirror wave activity were observed at the edges of the wake. Here the competition between the two instabilities is largely modified by the presence of

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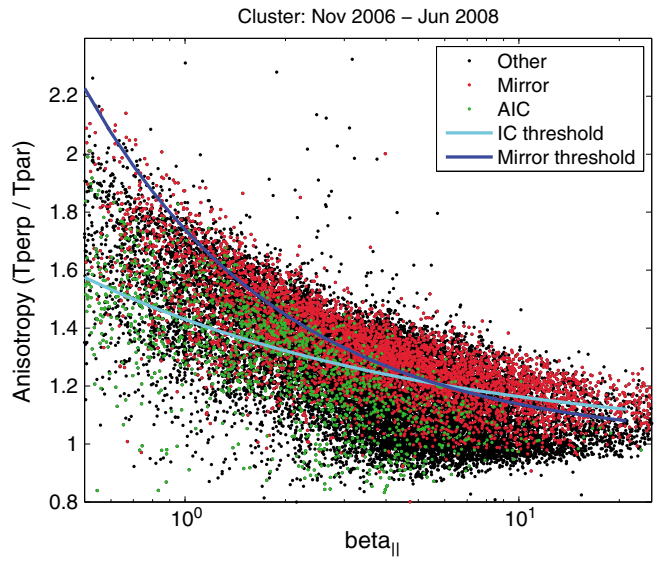

Figure 8. Distribution of observed wave types in the anisotropy-$\beta_{\|}$ space. Red dots indicate intervals with identified mirror modes, green dots with AIC waves, and black dots the remaining intervals in the statistical data set. Overplotted are the instability thresholds for mirror and ion cyclotron instabilities.

multiple ion species, and it was shown that the growth rate of mirror instability can exceed that of ion cyclotron waves at the edges of the wake due to a local increase in the relative density of the SO_2^+ pickup ions [*Huddleston et al.*, 1999]. In the studied case of terrestrial magnetosheath the minor ions (mostly He^{++}) are known to affect the competition of the two modes [*Gary et al.*, 1993], enabling the mirror waves to grow at lower values of ion β, but they do not seem to play as dominant role as in the case of Io wake. Similar to our results, it was reported [*Russell et al.*, 1999; *Pokhotelov et al.*, 2001] that the growth of mirror structures inhibits the ion cyclotron instability and both types of waves are not observed simultaneously. In these studies, it was postulated that the magnetic field gradients associated with mirror structures perturb the gyromotion of the ions and remove the ions from the cyclotron resonance. This mechanism provides an alternative explanation of the effect observed in our data and is consistent with the simulations of *Shoji et al.* [2009].

It should be noted that our data analysis technique does not identify mixed mirror-AIC events. This statistics might thus be biased toward mirror events due to their larger amplitude. From our experience and previous publications [e.g., *Anderson and Fuselier*, 1993] it is known that clear mixed AIC-mirror events exist but are not very frequently observed (under 10% in *Anderson and Fuselier* [1993] data set).

4. Conclusions

In this paper we have studied a statistical distribution of plasma parameters relevant to mirror and ion cyclotron instabilities as well as the distribution of mirror and AIC waves themselves in the terrestrial magnetosheath. Magnetosheath flow model [*Soucek and Escoubet*, 2012] was employed in the calculation of the relative location of the observation with respect to magnetosheath boundaries. The following main conclusions were reached from the statistical data:

1. Values of A, R, and $\beta_{\|}$ depend strongly on the parameters of the upstream shock crossing (Figure 4). High Mach number Q⊥ shocks result in the largest value of R, low Mach number Q⊥ shocks correspond to low values of $\beta_{\|}$ and high anisotropy. Highest values of $\beta_{\|}$ and low anisotropy are associated with high Mach number $Q\|$ shocks.
2. Strongest temperature anisotropy A and very low $\beta_{\|}$ are found in the subsolar region near the magnetopause where PDL is formed. In the flank magnetosheath, $\beta_{\|}$ low is correlated with moderate anisotropy. The most mirror unstable plasma is found in subsolar magnetosheath close to the bow shock.
3. Mirror waves occur in Q⊥ magnetosheath and are much more common in correlation with high Mach number shock. AIC waves also prefer Q⊥ shock conditions but appear predominantly during low Mach number conditions (Figure 5). Taking into account the above point and the conclusion of section 3.3, this can be

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seen as the consequence of the low-$\beta_{\|}$ high-anisotropy plasma with a low value of R present for Q⊥ low M_A conditions.

4. Both mirror and AIC waves are rarely observed in Q|| magnetosheath, consistent with previous results of *Anderson and Fuselier* [1993].
5. The occurrence rates of mirror waves are differently distributed for peaks and dips. Peaks strictly follow the distribution of R parameter, consistent with the results of *Soucek et al.* [2008] and *Génot et al.* [2009b] and appear in middle Q⊥ magnetosheath, consistent with *Verigin et al.* [2006] and *Génot et al.* [2009a]. Dips are observed in the flank magnetosheath and near the magnetopause - areas characterized by low $\beta_{\|}$.
6. AIC waves are observed behind the shock [*Song et al.*, 1994], in the magnetosheath flanks, and to lesser extent near subsolar magnetopause where PDL is formed [*Anderson and Fuselier*, 1993].
7. Both mirror and AIC waves follow their respective thresholds, but AIC waves were detected only below the mirror threshold. We interpret this effect as the result of a different mode of saturation of AIC and mirror instabilities in the nonlinear stage as described by *Shoji et al.* [2009]. After the instability reduces the plasma to a stable state, AIC waves are damped after a short time, while mirror instability saturates in the form of persistent oscillatory structures in the magnetic field which are much more likely to be observed.

This work presents comprehensive quantitative results on the distribution of ULF waves and ion parameters in the magnetosheath. Our results generally agree with partial results on specific dependencies published previously. The detection of waves performed in this article was adequate for the objective of this study, but the technique can be refined and the data set can be employed to study the detailed ULF wave properties within the magnetosheath. An investigation of analogous plasma parameters corresponding to He^{++} ions would also contribute to the understanding of magnetosheath ULF wave generation.

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